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(54) **POWER SUPPLY CONTROL APPARATUS AND SWITCHING POWER SUPPLY INCLUDING THE SAME**

(71) Applicant: **ROHM CO., LTD.**, Kyoto (JP)

(72) Inventors: **Tadashi Akaho**, Kyoto (JP); **Yasuto Yoshioka**, Kyoto (JP)

(73) Assignee: **Rohm Co., Ltd.**, Kyoto (JP)

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H02M 1/00 (2006.01)

(52) **U.S. Cl.**
CPC ***H02M 3/1588*** (2013.01); ***H02M 1/0032*** (2021.05)

(58) **Field of Classification Search**
CPC H02M 3/1588; H02M 1/0032
See application file for complete search history.

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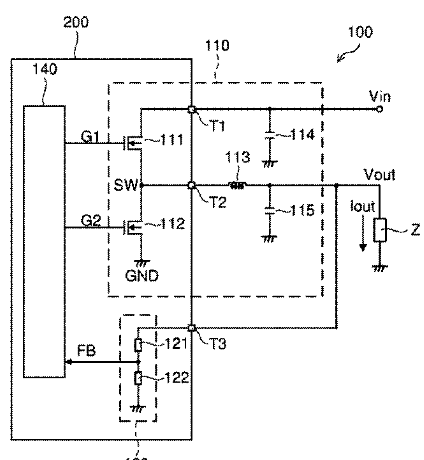
Primary Examiner — Kyle J Moody
(74) *Attorney, Agent, or Firm* — Fish & Richardson P.C.

(57) **ABSTRACT**

A power supply control apparatus, includes: a driver configured to respectively drive an output transistor and a synchronous rectification transistor configured to generate an output voltage from an input voltage and supply the output voltage to a load; and a controller configured to, in a light load mode in which output feedback control is performed such that a switching frequency of the output transistor becomes lower as the load becomes lighter, during an off period from a time at which both the output transistor and the synchronous rectification transistor are turned off to an on timing of the output transistor based on the output feedback control, periodically turn on the synchronous rectification transistor within a range in which the switching frequency does not fall below a predetermined lower limit value.

6 Claims, 3 Drawing Sheets



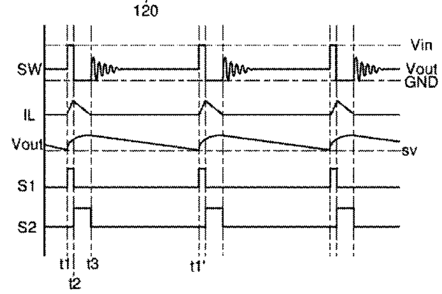


FIG. 1

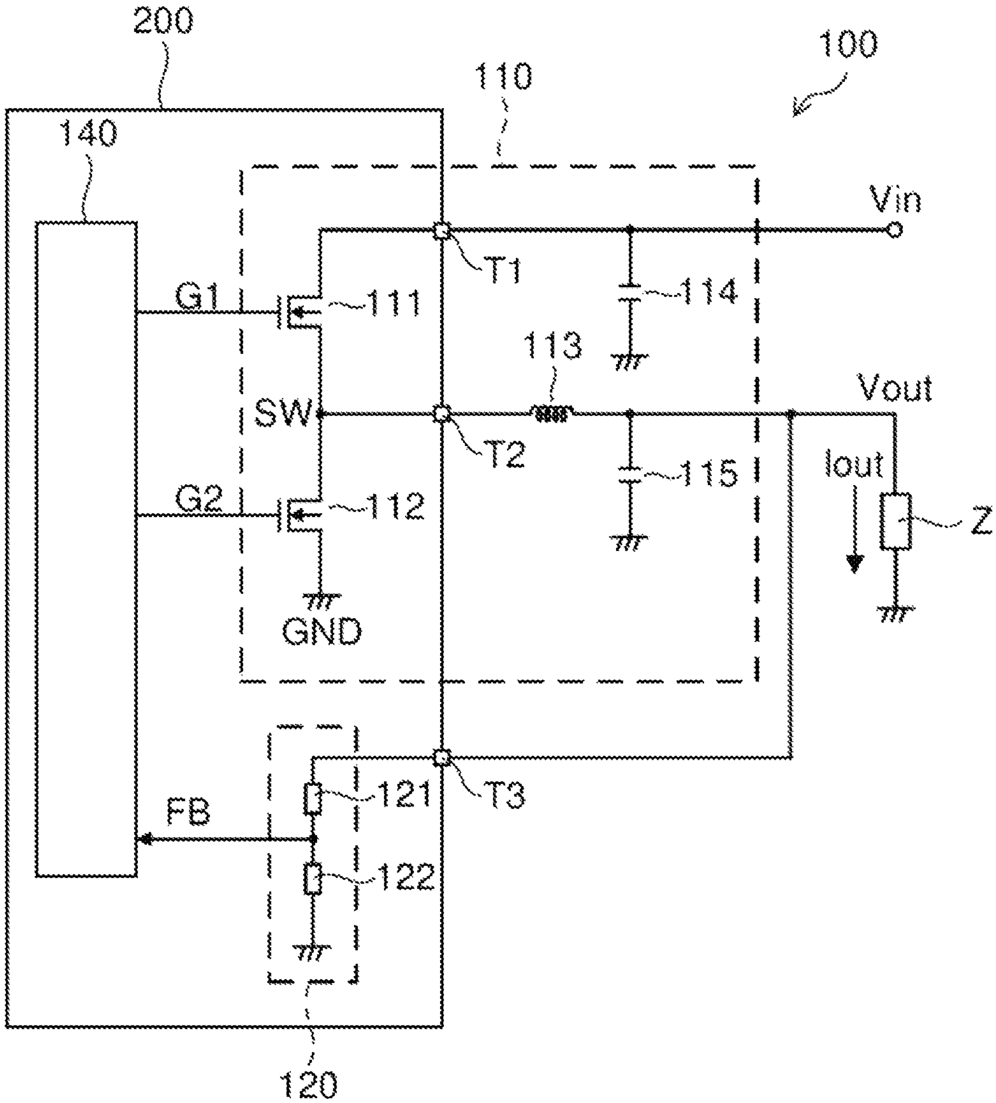

200
100
110
140
Vin
T1
114
G1
111
113
Vout
SW
T2
115
Iout
Z
G2
112
GND
T3
121
FB
122
120

FIG. 2

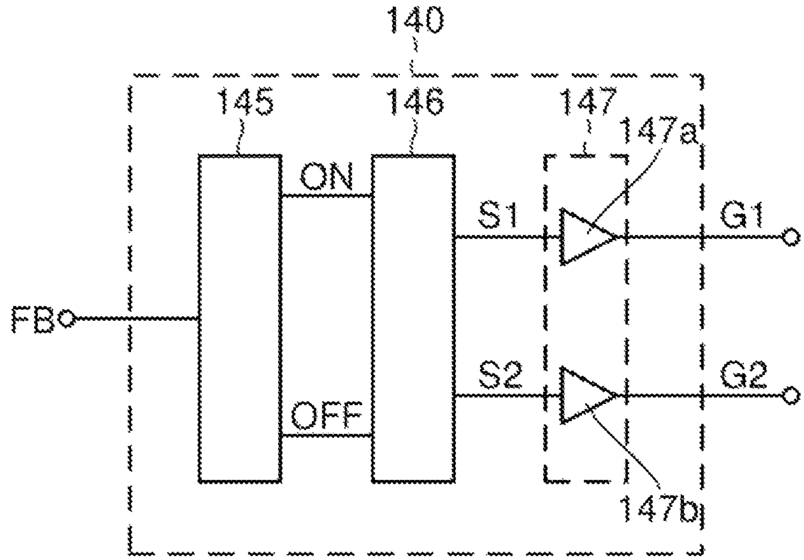

140
145
146
147
147a
ON
OFF
S1
S2
G1
G2
FB
147b

FIG. 3

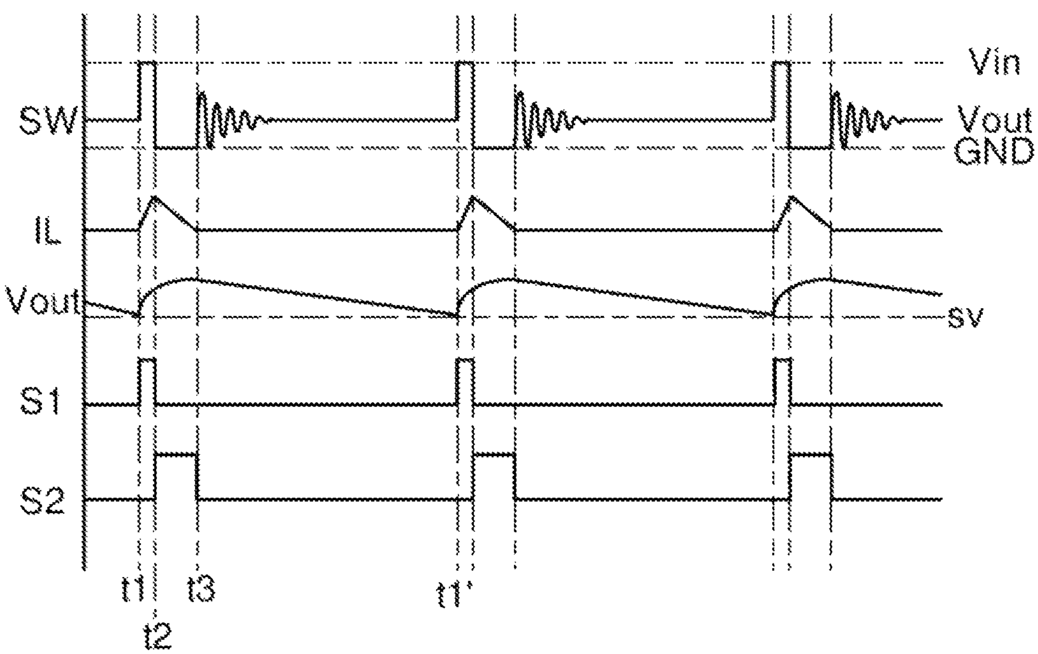

SW
IL
Vout
S1
S2
Vin
Vout
GND
sv
t1
t2
t3
t1'

FIG. 4

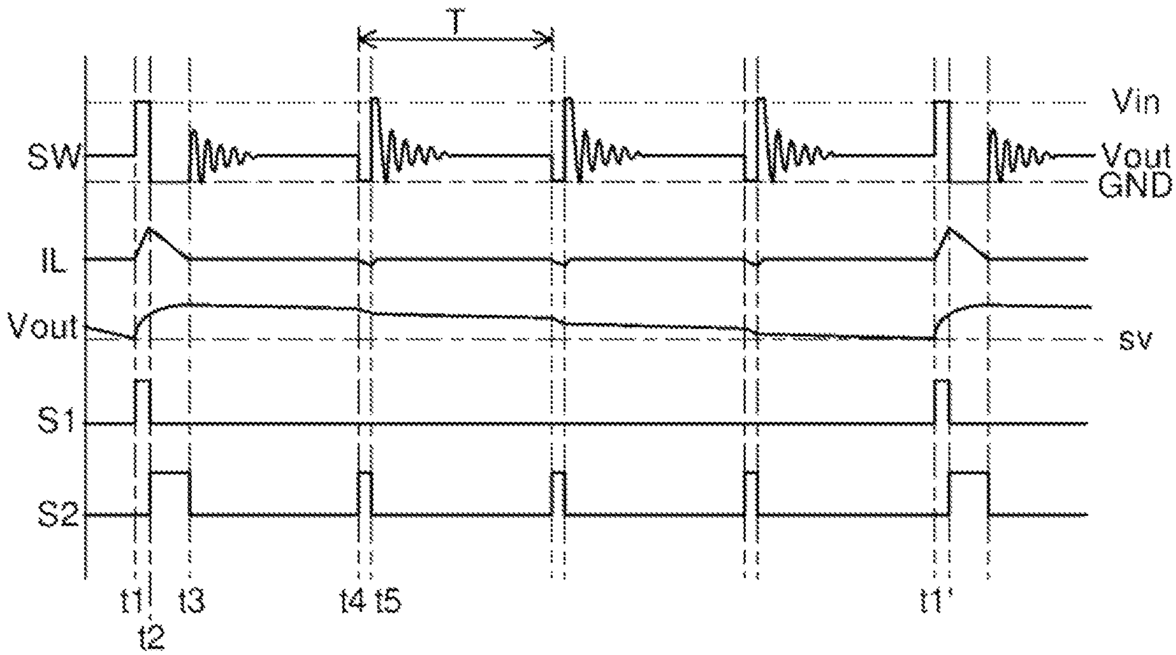
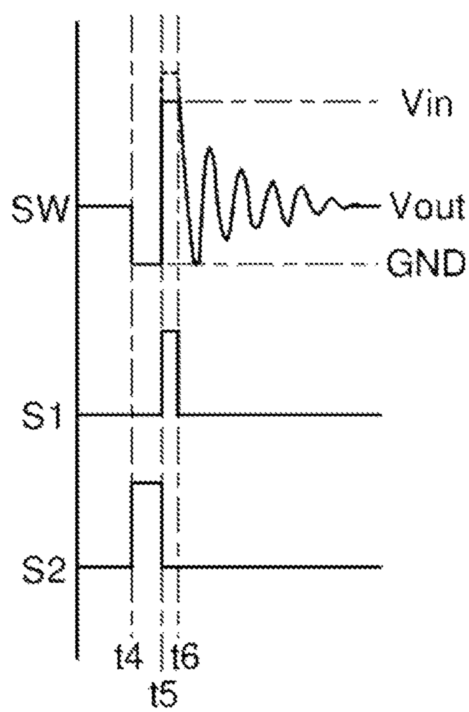

T
Vin
SW
Vout
GND
IL
Vout
sv
S1
S2
t1
t2
t3
t4
t5
t1'

FIG. 5


Vin
SW
Vout
GND
S1
S2
t4
t5
t6

POWER SUPPLY CONTROL APPARATUS AND SWITCHING POWER SUPPLY INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-156153, filed on Sep. 29, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to a power supply control apparatus and a switching power supply provided with the same.

BACKGROUND

In the related art, among switching power supplies, there is a switching power supply that has an operation mode (so-called light load mode) in which switching pulses are thinned out at a light load to reduce switching loss. In the related art, during the light load mode, a switching frequency fluctuates according to a load current. Therefore, depending on an amount of load current, the switching frequency may drop to a human audible range (generally 20 kHz or less), and input and output capacitors may generate an offensive sound (so-called switching power supply noise).

As a method of preventing a noise in a switching power supply, for example, it is conceivable that when a noise prevention function is turned on, a load resistor provided inside a power supply control IC is connected to a switch output stage to increase the load current and intentionally raise the switching frequency.

In the related art, as such a switching power supply, there is known a switching power supply including a load resistor circuit. The load resistor circuit of this switching power supply is provided inside the power supply control IC and connected to the switch output stage. The load resistor circuit is configured to detect a feedback voltage (a divided voltage of the output voltage generated by the switch output stage) and change the load current according to a value of the feedback voltage.

BRIEF DESCRIPTION OF DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the present disclosure.

FIG. **1** is a diagram showing an embodiment of a switching power supply.

FIG. **2** is a diagram showing a configuration example of a control circuit.

FIG. **3** is a timing chart showing a switching operation in a light load mode.

FIG. **4** is a timing chart showing a switching operation of a switch output stage in a silent light load mode.

FIG. **5** is an enlarged view of a part of a timing chart showing a modification of the switching operation of the switch output stage in the silent light load mode.

DETAILED DESCRIPTION

Reference will now be made in detail to various embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. However, it will be apparent to one of ordinary skill in the art that the present disclosure may be practiced without these specific details. In other instances, well-known methods, procedures, systems, and components have not been described in detail so as not to unnecessarily obscure aspects of the various embodiments.

Embodiments of the present disclosure will be described below with reference to the drawings.

<Switching Power Supply>

FIG. **1** is a diagram showing an embodiment of a switching power supply. The switching power supply **100** of the present embodiment is a DC/DC converter configured to generate a desired output voltage Vout from an input voltage Vin and supply the output voltage Vout to a load Z. The switching power supply **100** includes a control circuit **140** (power supply control apparatus), a switch output stage **110**, and a feedback voltage generation circuit **120**.

Except for some components (in this figure, an inductor **113** and capacitors **114** and **115**) included in the switch output stage **110**, the above-described components may be integrated into a semiconductor device **200** (so-called power supply control IC) configured to mainly control the switching power supply **100**. The semiconductor device **200** may appropriately incorporate arbitrary components (such as various protection circuits) other than those described above. The semiconductor device **200** also includes a plurality of external terminals T**1** to T**3** as means configured to establish electrical connection to the outside of the device.

The switch output stage **110** is a step-down type switch output stage configured to drive an inductor current IL to generate a desired output voltage Vout from an input voltage Vin by turning on and off upper and lower switches connected to form a half bridge. The switch output stage **110** includes an output transistor **111**, a synchronous rectification transistor **112**, an inductor **113**, and capacitors **114** and **115**.

The output transistor **111** is an NMOSFET [N-channel type metal oxide semiconductor field effect transistor]. The output transistor **111** functions as an upper switch of the switch output stage **110**.

Inside the semiconductor device **200**, a drain of the output transistor **111** is connected to an external terminal T**1** (a terminal to which an input voltage Vin is applied). A source of the output transistor **111** is connected to an external terminal T**2** (a terminal to which a switch voltage SW is applied). A gate of the output transistor **111** is connected to a terminal to which an upper gate signal G**1** is applied.

The output transistor **111** is turned on when the upper gate signal G**1** is at a high level, and is turned off when the upper gate signal G**1** is at a low level. When an NMOSFET is used as the output transistor **111**, a bootstrap circuit or a charge pump circuit (not shown in this figure) is required to raise the high level of the upper gate signal G**1** to a voltage value higher than the input voltage Vin.

The synchronous rectification transistor **112** is an NMOSFET. The synchronous rectification transistor **112** functions as a lower switch of the switch output stage **110**.

Inside the semiconductor device **200**, the drain of the synchronous rectification transistor **112** is connected to the external terminal T**2** (the terminal to which the switch voltage SW is applied). The source of the synchronous rectification transistor **112** is connected to a ground terminal (a terminal to which a ground voltage GND is applied). The gate of the synchronous rectification transistor **112** is connected to a terminal to which a lower gate signal G**2** is applied.

The synchronous rectification transistor **112** is turned on when the lower gate signal G**2** is at a high level, and is turned off when the lower gate signal G**2** is at a low level.

The inductor **113** and the capacitors **114** and **115** are discrete components externally attached to the semiconductor device **200**. A first end of the capacitor **114** is connected to the external terminal T**1** of the semiconductor device **200**. A second end of the capacitor **114** is connected to the ground terminal.

A first end of the inductor **113** is connected to the external terminal T**2** of the semiconductor device **200**. A second end of the inductor **113** and a first end of the capacitor **115** are connected to the terminal to which the output voltage Vout is applied and an external terminal T**3** of the semiconductor device **200**.

A second end of the capacitor **115** is connected to the ground terminal. The capacitor **114** functions as an input capacitor configured to smooth the input voltage Vin. Further, the inductor **113** and the capacitor **115** function as an LC filter configured to rectify and smooth the switch voltage SW to generate the output voltage Vout.

The output transistor **111** and the synchronous rectification transistor **112** are basically turned on and off in a complementary manner according to the upper gate signal G**1** and the lower gate signal G**2**. By such an on/off operation, a square-wave switch voltage SW pulse-driven between the input voltage Vin and the ground voltage GND is generated at the first end of the inductor **113**. The word "complementary" mentioned above should be understood as including not only a case where the on/off states of the output transistor **111** and the synchronous rectification transistor **112** are completely reversed, but also a case where there is provided a period (dead time) in which the output transistor **111** and the synchronous rectification transistor **112** are simultaneously turned off. Further, during a light load mode described later, both the output transistor **111** and the synchronous rectification transistor **112** are turned off, and the driving of the switch output stage **110** may be temporarily stopped (details of which will be described later).

Further, the output transistor **111** may be replaced with a PMOSFET. In such a case, the aforementioned bootstrap circuit and charge pump circuit become unnecessary.

It is also possible to externally attach the output transistor **111** and the synchronous rectification transistor **112** to the semiconductor device **200**. In such a case, instead of the external terminal T**2**, external terminals configured to output the upper gate signal G**1** and the lower gate signal G**2** to the outside of the device are required.

Further, when a high voltage is applied to the switch output stage **110**, a high-voltage element such as a power MOSFET, an IGBT (insulated gate bipolar transistor), and a SiC transistor may be used as the output transistor **111** or the synchronous rectification transistor **112**.

The feedback voltage generation circuit **120** includes resistors **121** and **122** connected in series between the external terminal T**3** (a terminal to which the output voltage Vout is applied) and the ground terminal. A feedback voltage FB (divided voltage of the output voltage Vout) corresponding to the output voltage Vout is outputted from a connection node between the resistors **121** and **122**.

When the output voltage Vout falls within an input dynamic range of the control circuit **140**, the feedback voltage generation circuit **120** may be omitted and the output voltage Vout itself may be directly inputted to the control circuit **140** as the feedback voltage FB.

<Control Circuit>

As basic output feedback control, the control circuit **140** performs a pulse width modulation control (PWM control) of the upper gate signal G**1** and the lower gate signal G**2** such that the feedback voltage FB matches a predetermined target value.

The control circuit **140** also has a light load mode (PFM (pulse frequency modulation) control) in which a switching loss is reduced by repeatedly stopping and restoring the drive of the switch output stage **110** within a range in which the output voltage Vout does not fall below the target value.

Further, the control circuit **140** may perform a silent light load mode during the light load mode. The silent light load mode is a function that periodically turns on the synchronous rectification transistor **112** such that the switching frequency does not fall below a lower limit value in the light load mode (details of which will be described later). The lower limit value is a frequency at which the switching power supply **100** does not generate a noise, and is, for example, about 21 to 25 kHz, which is higher than the human audible band.

FIG. **2** is a diagram showing a configuration example of the control circuit **140**. As shown in FIG. **2**, the control circuit **140** includes a signal controller **145** (controller), a logic circuit **146**, and a drive circuit (driver) **147**.

The signal controller **145** transmits an on signal ON (clock signal) and an off signal OFF to the logic circuit **146** according to the magnitude of the feedback voltage FB. The logic circuit **146** basically generates an upper control signal S**1** and a lower control signal S**2** according to the on signal ON and the off signal OFF.

The drive circuit **147** includes an upper driver **147***a* and a lower driver **147***b*. The upper driver **147***a* receives the input of the upper control signal S**1** and generates the upper gate signal G**1**. The lower driver **147***b* receives the input of the lower control signal S**2** and generates the lower gate signal G**2**. A buffer or an inverter may be used as the upper driver **147***a* and the lower driver **147***b*, respectively.

When the signal controller **145** generates the on signal ON, the logic circuit **146** raises the upper control signal S**1** to a high level and lowers the lower control signal S**2** to a low level. As a result, the output transistor **111** is turned on and the synchronous rectification transistor **112** is turned off. Then, the switch voltage SW rises to a high level (=Vin). This state is called a first phase.

On the other hand, when the signal controller **145** generates the off signal OFF, the logic circuit **146** lowers the upper control signal S**1** to a low level and raises the lower control signal S**2** to a high level. As a result, the output transistor **111** is turned off and the synchronous rectification transistor **112** is turned on. Then, the switch voltage SW falls to a low level (≈GND). This state is called a second phase.

In the pulse width modulation control, the control circuit **140** controls the switch output stage **110** such that the above-described first phase and second phase alternately continue. Therefore, the on time of the output transistor **111** (the high level period of the switch voltage SW) is controlled such that the on time becomes longer as the pulse generation timing of the off signal OFF becomes delayed, and the on time becomes shorter as the pulse generation timing of the off signal OFF becomes faster.

On the other hand, in the light load mode, the control circuit **140** controls the switch output stage **110** so as to include a third phase in addition to the first and second phases described above. In the third phase, the output transistor **111** and the synchronous rectification transistor **112** are turned off, such that the switch output stage **110** enters into a non-driving state (switch voltage SW-output voltage Vout).

Operation Example in Light Load Mode

FIG. **3** is a timing chart showing the switching operation in the light load mode. In FIG. **3**, the switch voltage SW, the inductor current IL, the output voltage Vout, the upper control signal **S1**, and the lower control signal **S2** are depicted in order from the top. The control circuit **140** executes the light load mode when the load current Iout flowing through the load Z falls below a predetermined value. The light load mode will be specifically described below with reference to FIG. **3**.

When the output voltage Vout drops to a reference voltage sv (time **t1** in FIG. **3**), the logic circuit **146** raises the upper control signal S1 to a high level and lowers the lower control signal S2 to a low level. As a result, the state of the first phase described above is entered, and the output voltage Vout rises.

When the on-time of the output transistor **111** elapses for a predetermined time (time **t2** in FIG. **3**), the logic circuit **146** lowers the upper control signal S1 to a low level and raises the lower control signal S2 to a high level. As a result, the state of the second phase described above is entered.

When the inductor current IL becomes 0 or negative (time **t3** in FIG. **3**), the logic circuit **146** lowers the upper control signal **S1** and the lower control signal **S2** to a low level. As a result, the state of the third phase described above is entered. Thereafter, the output voltage Vout gradually decreases with a slope corresponding to the load current Iout flowing through the load Z.

When the output voltage Vout drops to the reference voltage sv again (time **t1'** in FIG. **3**), the logic circuit **146** raises the upper control signal **S1** to a high level and keeps the lower control signal **S2** at a low level. As a result, the state of the first phase described above is entered again. Likewise, the control circuit **140** controls the switch output stage **110** in the light load mode such that the states of the first to third phases are repeated in the above-described order.

In the light load mode, the period from the time (time **t3** in FIGS. **3** and **4**) when both the output transistor **111** and the synchronous rectification transistor **112** are turned off to the next turn-on timing of the output transistor **111** (time **t1'** in FIGS. **3** and **4**) is called an "off period." In the silent light load mode described later, the output transistor **111** may turn off and the synchronous rectification transistor **112** may turn on during the off period.

When a time interval between the first phase and the next first phase (interval between time **t1** and time **t1'** in FIG. **3**) exceeds a predetermined time in the light load mode, i.e., when the switching frequency falls below the lower limit value, the control circuit **140** makes the silent light load mode valid.

Operation Example of Silent Light Load Mode

FIG. **4** is a timing chart showing the switching operation of the switch output stage **110** in the silent light load mode. The silent light load mode will be specifically described below with reference to this figure.

As described above, the control circuit **140** controls the switch output stage **110** to sequentially transition to the first phase, the second phase, and the third phase in the light load mode. When the silent light load mode is made valid, the control circuit **140** periodically turns on the synchronous rectification transistor **112** during the off period (period from time **t3** to time **t1'** in FIG. **4**).

More specifically, in the state of the third phase (time **t3** in FIG. **4**), the logic circuit **146** keeps the upper control signal S1 at a low level and raises the lower control signal S2 to a high level at a predetermined timing (time **t4** in FIG. **4**). As a result, the synchronous rectification transistor **112** is turned on while the output transistor **111** is kept turned off. The on time of the synchronous rectification transistor **112** in this case (time period from time **t4** to time **t5** in FIG. **4**) is shorter than the on time of the synchronous rectification transistor **112** in the second phase (time period from time **t2** to time **t3** in FIG. **4**).

When the on time of the synchronous rectification transistor **112** elapses (time **t5** in FIG. **4**), the logic circuit **146** causes the lower control signal **S2** to fall to a low level while keeping the upper control signal **S1** at a low level. As a result, both the output transistor **111** and the synchronous rectification transistor **112** are turned off again.

At this time (time **t5** in FIG. **4**), the synchronous rectification transistor **112** is switched from on to off while the output transistor **111** kept turned off, thereby generating a counter electromotive force. More specifically, during the period from time **t4** to time **t5**, an inductor current IL flows from the inductor **113** to the ground terminal via the synchronous rectification transistor **112**. That is, the inductor current IL falls below a zero value. When the synchronous rectification transistor **112** is turned off, the inductor **113** attempts to keep the inductor current IL continuously flowing in the same direction as thus far (i.e., in the negative direction) due to the counter electromotive force generated in itself. As a result, a polarity of the switch voltage SW is switched from negative to positive.

In such a state, the charges stored in the capacitor **115** are returned to the input side via the inductor **113**. Since the output transistor **111** is turned off at this time, the charges are returned to the input side via a body diode built in the output transistor **111**.

The magnitude of the counter electromotive force is changed according to the on time of the synchronous rectification transistor **112** in this case (the time period from time **t4** to time **t5**). More specifically, as the on time of the synchronous rectification transistor **112** becomes longer, the counter electromotive force becomes larger, and as the on time becomes shorter, the counter electromotive force becomes smaller.

The logic circuit **146** changes the on time (time period from time **t4** to time **t5** in FIG. **4**) of the synchronous rectification transistor **112** during the off period within a range where the switch voltage SW rises close to the vicinity of the high level and the output voltage Vout does not fall below the reference voltage sv. The vicinity of the high level refers to a value at least within a range where the rise in the switch voltage SW affects the switching frequency.

Even after time **t5**, the logic circuit **146** periodically turns the synchronous rectification transistor **112** on and off until the turn-on timing (time **t1'** in FIG. **3**) of the output transistor **111** based on the output feedback control is reached. For example, when the switching frequency is 25 kHz, the synchronous rectification transistor **112** is turned on and off by setting one period T to 40 μs.

During this time, the output transistor **111** is kept turned off. By periodically turning on the synchronous rectification transistor **112** during the off period as described above, the drive period of the switch output stage **110** is shortened and the switching frequency is increased.

As described above, in the light load mode, the logic circuit **146** periodically turns on the synchronous rectification transistor **112** during the off period. Therefore, it is possible to prevent the switching frequency from falling below the predetermined lower limit. Accordingly, by setting the lower limit value to be higher than the human audible band, it is possible to suppress the noise of the switching power supply **100**.

In addition, in the silent light load mode, the on time of the synchronous rectification transistor **112** (the time period from time t**4** to time t**5** in FIG. **4**) is relatively short, and the output voltage Vout does not fall below the reference voltage sv. Therefore, it is possible to prevent a reduction in efficiency by executing the silent light load mode and turning on the synchronous rectification transistor **112**.

MODIFICATION

In addition to the above-described embodiments, various technical features described in the present disclosure may be modified in various ways without departing from the gist of the technical features. That is, the above-described embodiments should be considered as being exemplary and not restrictive in all respects. The technical scope of the present disclosure is not limited to the above-described embodiments and should be understood to include all changes falling within the meaning and range equivalent to the claims.

For example, mutual replacement of bipolar transistors with MOS field effect transistors and logic level inversion of various signals are optional.

Further, for example, in the above-described embodiments, in the silent light load mode, the output transistor **111** is kept turned off at the timing of turning off the synchronous rectification transistor **112** during the off period, and the switch voltage SW is raised to the vicinity of the high level by the counter electromotive force. However, the following configuration may be adopted.

FIG. **5** is an enlarged view of a part of the timing chart showing a modification of the switching operation of the switch output stage **110** in the silent light load mode. As shown in FIG. **5**, in the silent light load mode, the logic circuit **146** raises the lower control signal S**2** to a high level while keeping the upper control signal S**1** at a low level during the off period (time t**4** in FIG. **5**). Thereafter, the lower control signal S**2** is lowered to a low level and the upper control signal S**1** is raised to a high level (time t**5** in FIG. **5**). As a result, the synchronous rectification transistor **112** is turned on while keeping the output transistor **111** turned off. Then, the output transistor **111** is turned on and the synchronous rectification transistor **112** is turned off. When the output transistor **111** is turned on and the synchronous rectification transistor **112** is turned off (between time t**5** and time t**6** in FIG. **5**), the switch voltage SW becomes equal to the input voltage Vin and is lower than the raised value due to the above-described counter electromotive force (portion indicated by the dashed line in the graph of FIG. **5**).

The reason is that the current flowing back to the input side (input voltage Vin's side) flows directly from the source of the output transistor **111** to the input side through the drain of the output transistor **111** without passing through the body diode of the output transistor **111**. Then, a relatively large amount of current may flow through the input side without being affected by the voltage drop caused by the body diode. Therefore, by providing a rechargeable device such as a battery on the input side, it becomes possible to recover most of the electric charges stored in the capacitor **115**, which makes it possible to improve the efficiency.

In this modification, as shown in FIG. **5**, when the synchronous rectification transistor **112** is turned off, the output transistor **111** is turned on at the same time. Alternatively, the control circuit **140** may be configured to turn on the output transistor **111** when it detects that the synchronous rectification transistor **112** is turned off.

The power supply control apparatus (**140**) disclosed in the present disclosure includes: a driver (**147**) configured to respectively drive an output transistor (**111**) and a synchronous rectification transistor (**112**) configured to generate an output voltage (Vout) from an input voltage (Vin) and supply the output voltage (Vout) to a load (Z); and a controller (**145**) configured to, in a light load mode in which output feedback control is performed such that a switching frequency of the output transistor (**111**) becomes lower as the load (Z) becomes lighter, during an off period from a time at which both the output transistor (**111**) and the synchronous rectification transistor (**112**) are turned off to an on timing of the output transistor (**111**) based on the output feedback control, periodically turn on the synchronous rectification transistor (**112**) within a range in which a switching frequency does not fall below a predetermined lower limit value (First Configuration).

In the power supply control apparatus (**140**) of the First Configuration, the on timing of the output transistor (**111**) based on the output feedback control may be a time at which the output voltage (Vout) drops to a reference voltage (sv), and the controller (**145**) may be further configured to change an on time of the synchronous rectification transistor (**112**) during the off period such that the output voltage (Vout) does not fall below the reference voltage (sv) due to the turn-on of the synchronous rectification transistor (**112**) during the off period (Second Configuration).

In the power supply control apparatus (**140**) of the First Configuration, the controller (**145**) may be further configured to turn off the synchronous rectification transistor (**112**) after turning on the synchronous rectification transistor (**112**) during the off period, and turn on the output transistor (**111**) in synchronization with the turning-off of the synchronous rectification transistor (**112**) (Third Configuration).

In the power supply control apparatus (**140**) of the First Configuration, the lower limit value may be 25 kHz (Fourth Configuration).

In the power supply control apparatus (**140**) of any one of the First to Fourth Configurations, the apparatus may be integrated in a semiconductor device (**200**) (Fifth Configuration).

The switching power supply (**100**) disclosed in the present disclosure includes the power supply control apparatus (**140**) of any one of the First to Fourth Configurations (Sixth Configuration).

According to the power supply control apparatus (**140**) of the First Configuration, it is possible to suppress the switching frequency from falling below the predetermined lower limit value in the light load mode. Therefore, by setting the lower limit value to a value higher than the human audible band, it is possible to suppress generation of a noise that may be recognized by humans. Further, since the power supply control apparatus (**140**) is not configured to involve changing a value of the load (Z) or forcibly discharge the charges accumulated in the output capacitor, it is possible to suppress a decrease in efficiency.

Further, according to the power supply control apparatus (**140**) of the Second Configuration, it is possible to suppress the output voltage (Vout) from falling below the reference voltage (sv) by turning on the synchronous rectification transistor (**112**) during the off period. Therefore, even when the load is light (Z), it is possible to prevent the on timing of the output transistor (**111**) from becoming short based on the output feedback control, thereby suppressing a decrease in efficiency.

Further, according to the power supply control apparatus (**140**) of the Third Configuration, the current flows directly to the side of the input voltage (Vin) via the output transistor (**111**) without going through the body diode of the output transistor (**111**). For this reason, the flowing current is not affected by a voltage drop due to the body diode. Therefore, it is possible to allow a larger amount of current to flow, and it is possible to suppress a decrease in efficiency when adopting a configuration in which a battery or the like is provided on the side of the input voltage (Vin).

Further, according to the power supply control apparatus (**140**) of the Fourth Configuration, it is possible to prevent the switching frequency from becoming lower than 25 kHz, which is higher than the general human audible band, and it is possible to suppress generation of a noise in the switching power supply (**100**).

Moreover, according to the switching power supply (**100**) of the Sixth Configuration, it is possible to suppress generation of a noise while suppressing a decrease in efficiency.

The power supply control apparatus disclosed in the present disclosure may be used as a main controller of switching power supplies installed in various applications, thereby suppressing generation of a noise in the switching power supplies and suppressing a decrease in efficiency of the switching power supplies.

While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosures. Indeed, the embodiments described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosures. The accompanying claims and their

equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosures.

What is claimed is:

1. A power supply control apparatus, comprising:

a driver configured to respectively drive an output transistor and a synchronous rectification transistor configured to generate an output voltage from an input voltage and supply the output voltage to a load; and

a controller configured to, in a light load mode in which output feedback control is performed such that a switching frequency of the output transistor becomes lower as the load becomes lighter, during an off period from a time at which both the output transistor and the synchronous rectification transistor are turned off to an on timing of the output transistor based on the output feedback control, periodically turn on the synchronous rectification transistor within a range in which the switching frequency does not fall below a predetermined lower limit value.

2. The power supply control apparatus of claim **1**, wherein the on timing of the output transistor based on the output feedback control is a time at which the output voltage drops to a reference voltage, and

wherein the controller is further configured to change an on time of the synchronous rectification transistor during the off period such that the output voltage does not fall below the reference voltage due to the turning-on of the synchronous rectification transistor during the off period.

3. The power supply control apparatus of claim **1**, wherein the controller is further configured to turn off the synchronous rectification transistor after turning on the synchronous rectification transistor during the off period, and turn on the output transistor in synchronization with the turning-off of the synchronous rectification transistor.

4. The power supply control apparatus of claim **1**, wherein the predetermined lower limit value is 25 kHz.

5. The power supply control apparatus of claim **1**, wherein the power supply control apparatus is integrated in a semiconductor device.

6. A switching power supply, comprising:

the power supply control apparatus of claim **1**.

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